

MODULE 04

Rheumatologic Arthropathies: Pattern-Based Radiographic Diagnosis

RA · PsA · Gout · CPPD · AS · Sacroiliitis · Six-Axis Framework

1.0 AMA PRA Cat 1

General MSK

CME Article Review

LEARNING OBJECTIVES

1. Apply a six-axis framework to systematically analyze arthropathy radiographs.
2. Identify distinguishing features of RA, PsA, gout, CPPD, and AS.
3. Recognize joint-specific patterns: pancarpal RA, pencil-in-cup PsA, overhanging edge gout.
4. Differentiate sacroiliitis patterns: symmetric (AS/IBD) vs. asymmetric (PsA/ReA).
5. Apply modified New York criteria to sacroiliitis grading (0–4).
6. Recognize hallmark deformities of RA and atlantoaxial subluxation implications.

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1. Six-Axis Framework for Arthropathy Analysis

Axis	Key Question	Discriminating Diagnoses
1. Distribution	Symmetric vs. asymmetric? Ray pattern? Axial?	Symmetric→RA, CPPD; Asymmetric→PsA, Gout; Ray→PsA
2. Bone Density	Periarticular osteopenia? Preserved?	Osteopenia→RA, JIA; Preserved→Gout, PsA, OA, CPPD
3. Erosion	Marginal? Central (seagull)? Overhanging edge?	Marginal→RA; Central→EOA; Overhanging→Gout
4. Joint Space	Uniform narrowing? Focal? Pan-compartmental?	Uniform→RA; Focal→OA; Pancarpal+uniform→RA
5. New Bone	Osteophytes? Syndesmophytes? Periostitis?	Osteophytes→OA; Syndesmophytes→AS; Periostitis→PsA
6. Soft Tissue	Tophi? Calcification type?	Tophi→Gout; Chondrocalcinosis→CPPD; Calcinosis→SSc

2. RA & Psoriatic Arthritis — Key Features

Rheumatoid Arthritis vs. Psoriatic Arthritis

Feature	RA	PsA
Distribution	Symmetric; MCP/PIP/wrist; DIP SPARED	Asymmetric; DIP involved; ray pattern
Bone density	PERIARTICULAR OSTEOPENIA (hallmark)	PRESERVED density — key differentiator
Erosion	Marginal at "bare area"; radial MCP heads first	Pencil-in-cup DIP; "mouse ear"; central erosion
Joint space	Pancarpal collapse; uniform narrowing	Preserved or eccentric; DIP may fuse
New bone	ABSENT (no osteophytes)	Periostitis; enthesophytes; fluffy periosteal reaction
SI joints	Not involved	ASYMMETRIC sacroiliitis
Spine	Step-ladder subaxial; atlantoaxial subluxation	Bulky non-marginal asymmetric ossification

3. Crystal Arthropathies & Sacroiliitis Grading

Crystal Arthropathy Rapid Comparison

Feature	CPPD	HADD (Apatite)	Gout (Urate)
Crystal	Calcium pyrophosphate	Calcium hydroxyapatite	Monosodium urate
Location	TFC (wrist), menisci (knee)	Rotator cuff; peritendinous	1st MTP, ankle, olecranon, wrist
Erosion	Mild; SLAC wrist	None	Overhanging edge; sclerotic margin
Joint space	Narrowed (patellofemoral knee; SLAC wrist)	Preserved	Preserved until late
Special	Crowned dens on CT; chondrocalcinosis	Acute calcific periarthritis	Tophi may calcify late; radiolucent early

Modified New York Criteria — Sacroiliitis Grading

Grade	Finding
0 — Normal	No changes
1 — Suspicious	Equivocal; not clearly sacroiliitis
2 — Minimal	Small erosions or sclerosis; no joint space change
3 — Moderate	Erosion + sclerosis ± widening/narrowing ± partial ankylosis
4 — Ankylosis	Complete bilateral fusion

KEY POINT

ASAS MRI active sacroiliitis: BME on STIR at the SI joint subchondral region in ≥ 2 consecutive slices, or ≥ 2 locations on one slice. Gadolinium enhancement is an alternative criterion.

References

- Resnick D, Niwayama G. Diagnosis of Bone and Joint Disorders. 4th ed. Saunders 2002.